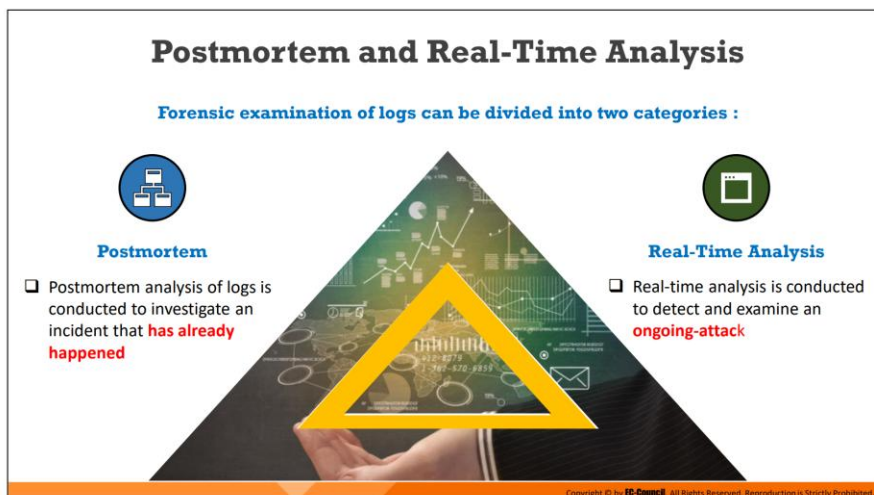
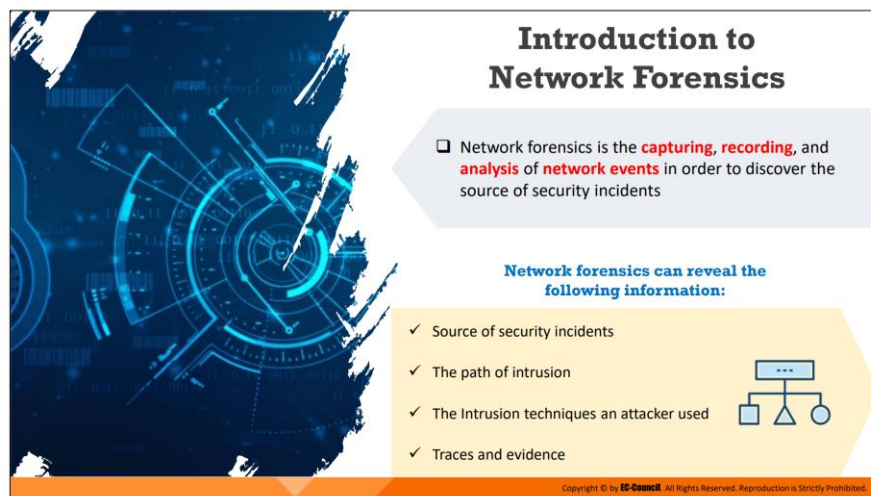
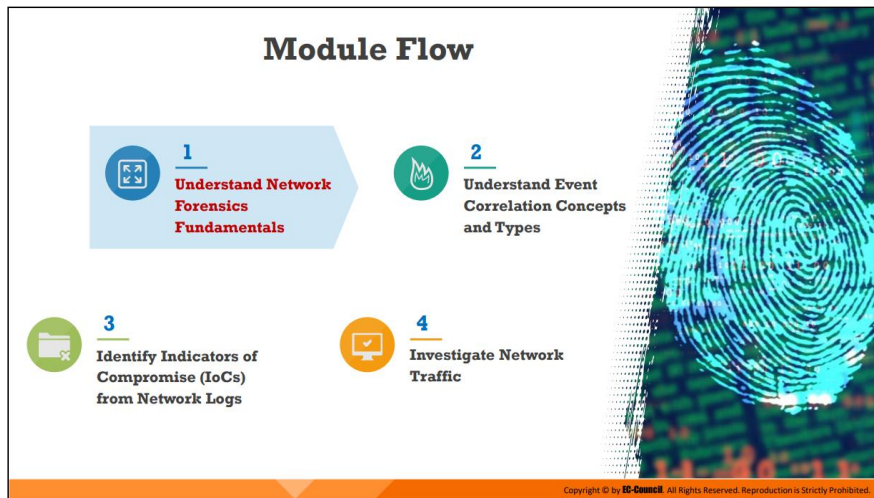


iNet Forensics

<https://chatgpt.com/share/680fdfad-c478-8010-becf-5e9a9306bc03>



Network Attacks

Most common attacks on networks

- Eavesdropping
- Data Modification
- IP Address Spoofing
- Denial-of-Service Attack
- Man-in-the-Middle Attack
- Packet Sniffing
- Enumeration
- Session Hijacking
- Buffer Overflow
- Email Infection
- Malware attack
- Password-based attack
- Router Attacks

Attacks specific to wireless networks

- Rogue Access Point Attack
- Client Misassociation
- Misconfigured Access Point Attack
- Unauthorized Association
- Ad Hoc Connection Attack
- Honeytrap Access Point Attack
- AP MAC Spoofing
- Jamming Attack

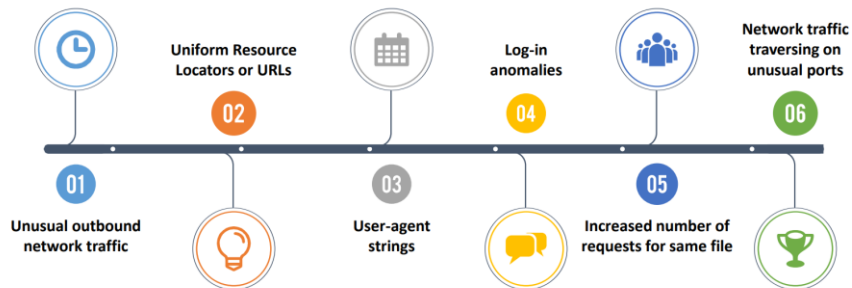


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Indicators of Compromise (IOCs)

- Indicators of Compromise (IOCs) are **digital forensic artifacts** that help detect a security incident that **has occurred** (or **is ongoing**) on a host system or a network
- These artifacts include logs related to systems, applications, networks, firewalls, etc.

Examples of IOCs:



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Where to Look for Evidence



Logs collected in **network devices** and **applications** can be used as evidence for investigating network security incidents

Functions	Layers of TCP/IP Model	Protocols	Network Devices and Applications Logs
Handles high-level protocols, issues of representation, encoding, and dialog control	Application layer	File Transfer (TFTP, FTP, NFS), Email (SMTP), Network Management (SNMP), Name Management (DNS)	Servers/Desktops, Anti-virus, Business Applications, Databases
Provides a logical connection between the endpoints and provides transport	Transport layer	Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)	Firewall, IDS/IPS
Selects the best path through the network for data flow	Internet layer	Internet Protocol (IP), Internet Control Message Protocol (ICMP), Address Resolution Protocol (ARP)	Firewall, IDS/IPS, VPN
Defines how to transmit an IP datagram to the other devices	Network Access layer	Ethernet, Fast Ethernet, SLIP, PPP, FDDI, ATM, Frame Relay, SMDs, ARP, Proxy ARP, RARP	Routers and Switches

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Types of Network-based Evidence

Full Content Data

- ❖ Full content data refers to **actual packets** that are collected by storing the network traffic (known as packet capture or PCAP files)
- ❖ Investigators can use tools like **tcpdump** or **Wireshark** to analyze any subset of full content data

Session Data

- ❖ Session data refers to a summary of **conversation** between **two network entities**
- ❖ It includes details such as destination IP and destination port, source IP and source port, start time of the session, and amount of information exchanged during the session

Alert Data

- ❖ Alert data is triggered by tools such as **Snort IDS** and **Suricata** that are preprogrammed to examine network traffic for IoCs and report the findings as alerts
- ❖ Investigators need to be careful while examining alert data as there might be **false positive** alerts

Statistical Data

- ❖ Statistical data provides overall profile or **summaries** of the **network traffic**
- ❖ Statistical data analysis can provide investigator with useful information such as **timestamps** related to network conversations, **protocols** and **services** being used, average **packet size**, and average **packet rate**

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Event Correlation

1

Event correlation is the process of relating a set of events that have occurred in a **predefined interval of time**

2

It includes analysis of events to determine how it could add up to become a bigger event

3

It is usually performed on a **log management platform**, after identifying logs having similar properties

Steps in event correlation:

Event aggregation

Event masking

Event filtering

Root cause analysis

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Prerequisites of Event Correlation

Transmission of Data

- ❑ Transmitting log data from one security device to another until it **reaches a consolidation point in the automated system**
- ❑ To have a secure transmission and reduce the risk of exposure during transmission, the data must be **encrypted and authenticated**

Normalization

- ❑ After the data is gathered, it must be **formatted** again from different log formats to a single or polymorphic log that can be easily inserted into the database

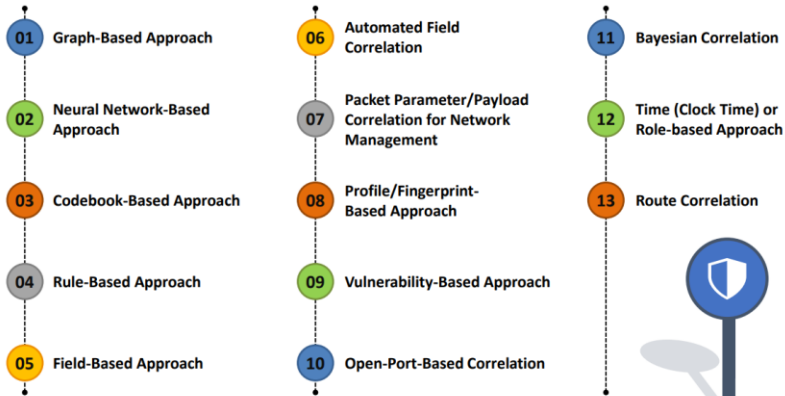


Data Reduction

- ❑ After collecting the data, repeated data must be **removed** so that the data can be correlated more efficiently
- ❑ Removing unnecessary data can be done by **compressing the data, deleting repeated data, filtering**, or combining similar events into a single event, and sending that to the correlation engine

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Event Correlation Approaches



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Analyzing Firewall Logs

- Firewalls are the **first points of entry into a network** and store details of all the data packets moving in and out of the network
- The network **firewall logs collect network traffic data** such as request source and destination, ports used, time and date, and priority
- These details help investigators **correlate the data with other suspicious files** to identify the source and other targets of an attack
- Investigators need to analyze the logs carefully based on the **timings and suspicious IP addresses**
- Check for **application generated requests, DNS queries, suspicious IP addresses, and URLs**



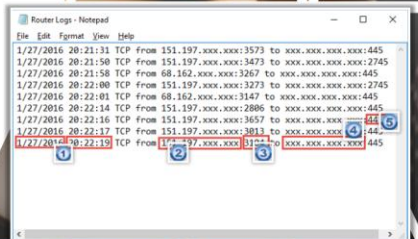
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Analyzing Router Logs

- Routers **store network connectivity logs** with details such as date, time, source and destination IPs, and ports used
- This information can help investigators in **verifying the timestamps** of an attack
- This also enables investigators to **correlate various events** to identify the **source and destination IP**
- Routers contain various **standards for storing the log** details of a network

The incoming log details are as follows:

1. Date and time
2. Source IP address
3. Source port
4. Destination IP address
5. Destination port



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Analyzing DHCP Logs

Sample DHCP Audit Log File

- ❑ DHCP logs are saved in the **C:\Windows\System32\dhcp** folder on DHCP servers

- ❑ **DHCP server log file format** is as follows:

Field	Description
ID	A DHCP Event ID code
Date	The date on which this entry was logged on the DHCP server
Time	The time at which this entry was logged on the DHCP server
Description	A description of this DHCP server event
IP Address	The IP address of the DHCP client
Host Name	The host name of the DHCP client
MAC Address	The media access control (MAC) address used by the network adapter hardware of the client

```
DhcpLog-Writing - Notepad
File Edit Format View Help
Microsoft DHCP Service Activity Log

Event ID Meaning
00 The log was started.
01 The log was stopped.
02 The log was temporarily paused due to low disk space.
10 A new IP address was leased to a client.
11 A lease was renewed by a client.
12 A lease was released by a client.
13 An IP address was found to be in use on the network.
14 A lease request could not be satisfied because the scope's address pool was exhausted.
15 A lease was denied.
16 A lease was deleted.
17 A lease was expired and DNS records for an expired lease have not been deleted.
18 A lease was expired and DNS records were deleted.
20 A BOOTP address was leased to a client.
21 A dynamic BOOTP address was leased to a client.
22 A BOOTP request could not be satisfied because the scope's address pool for BOOTP was exhausted.
23 A BOOTP IP address was deleted after checking to see it was not in use.
24 IP address cleanup operation has begun.
25 IP address cleanup statistics.
30 DNS update request to the named DNS server.
31 DNS update failed.
32 DNS update successful.
33 Packet dropped due to WAP policy.
34 DNS update request failed as the DNS update request queue limit exceeded.
35 DNS update request failed.
36 Packet dropped because the server is in failover standby role or the hash of the client ID does not match.
50+ Codes above 50 are used for Rogue Server Detection Information.

@Result: 0: NoQuarantine, 1:Quarantine, 2:Drop Packet, 3:Probation, 6:No Quarantine Information ProbationTime:Year-
ID,Date,Time,Description,IP Address,Host Name,MAC Address,User Name,TransactionID, @Result,ProbationTime, Correl
00,12/04/19,11:14:37,Started,,,,,R,S,,,,,,8
<
```

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Gathering Evidence via Sniffers

Sniffer is a computer software or hardware that can **intercept** and **log traffic** passing over a network

Sniffers put NICs in **promiscuous mode** to allow them to listen to and capture all data transmitted over the network

Sniffers **collect traffic** from the **network** and **transport layers** other than the physical and data-link layer

Investigators should **configure sniffers** for the size of frames to be captured



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Analyze Traffic for Sniffing Attempts

- 1 Attackers sniff the network traffic for **sensitive** information
- 2 **Different techniques** may be used for sniffing the traffic depending on the type of network
- 3 Passive sniffing is used to sniff a **hub-based** network while active sniffing is used to sniff a **switch-based** network
- 4 An attacker uses **MAC flooding** and **ARP poisoning** to sniff network traffic, which can then serve as a basis for launching **man-in-the middle (MITM) attacks**
- 5 **Identify sniffing attempts** by detecting the **signs** of MAC flooding and ARP poisoning using **Wireshark**

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Module Summary

- 1 This module has discussed the fundamentals of network forensics, including logging fundamentals
- 2 It also discussed in detail the concepts related to event correlation
- 3 Further, this module explained the identification of indicators of compromise (IoCs) from network logs
- 4 Finally, this module ended with a detailed discussion on investigating network traffic
- 5 In the next module, we will discuss the investigation of web attacks in detail



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Snort

- cs Network analysis tools are integral to keeping networks secure by providing real-time logging and analysis of events and traffic. Snort is atop this list of analysis tools as the most popular network intrusion detection and prevention system. The open-source version of the Snort IDS/IPS still has an active community of contributors and developers, while Cisco has developed commercial versions of Snort since acquiring parent company Sourcefire in 2013.
- cs Snort is largely used passively on the network, but it can also take action on malicious packets, making it a powerful detection tool for defenders. An attacker who could blind this tool to malicious traffic, however, could gain an important advantage over network defenders.
- cs Rules may trigger one of three actions: Alert rules: Generate an alert, Log rules: Alert and log the alert, Pass rules: Ignore the packet
- cs Some of the preprocessors included in Snort are: ARP, DNS, SSH and some OT (operational technology) protocols, such as MODBUS / DNP3.
- cs GET Request: alert tcp any -> any 80 (content:"GET "; content:"HTTP/1." sid:123456; rev:1;)
- cs FTP Bounce Attack: alert tcp any -> any 21 (msg:"FTP PORT bounce"; flow:to_server,established; content:"PORT"; nocase; ftpbounce; pcre:"/PORT/smi"; classtype:misc-attack; sid:123456789; rev:1;)
- cs SSH Brute Force Attack: drop tcp 10.1.2.100 any -> 10.1.1.100 22 (msg:"SSH Brute Force Attempt"; flow:established,to_server; content:"SSH"; nocase; offset:0; depth:4; detection_filter: track by_src, count 30, seconds 60; sid:1000001; rev:1;)

